

● MEDIUM

● GEOMETRY AND TRIGONOMETRY

Circles

5 questions · ~8 min

02

Q1

GEOMETRY AND TRIGONOMETRY

A circle in the xy -plane has the equation $(x - 13)^2 + (y - k)^2 = 64$. Which of the following gives the center of the circle and its radius?

- A. The center is at $(13, k)$ and the radius is 8.
- B. The center is at $(k, 13)$ and the radius is 8.
- C. The center is at $(k, 13)$ and the radius is 64.
- D. The center is at $(13, k)$ and the radius is 64.

Q2

GRID-IN

GEOMETRY AND TRIGONOMETRY

In the xy -plane, the graph of the equation $x - 3^2 + y - 5^2 = 9$ is a circle. The point $6, c$, where c is a constant, lies on this circle. What is the value of c ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

$$x^2 + 58x + y^2 = 0$$

In the xy -plane, the graph of the given equation is a circle. What are the coordinates x, y of the center of the circle?

- A. 0, 29
- B. 0, - 29
- C. 29, 0
- D. -29, 0

Q4

GEOMETRY AND TRIGONOMETRY

The measure of angle R is $\frac{2\pi}{3}$ radians. The measure of angle T is $\frac{5\pi}{12}$ radians greater than the measure of angle R . What is the measure of angle T , in degrees?

- A. 75
- B. 120
- C. 195
- D. 390

What is the center of the circle in the xy -plane defined by the equation $x - 1^2 + y + 7^2 = 1$?

- A. -1, -7
- B. -1, 7
- C. 1, -7
- D. 1, 7